

Lab 12 Pair Programming – Recursion

Aim

This week lab is focused on applying using Recursion to solve problems.

在计算机实验室学习编程时，培养计算思维技能的一种有效方法是采用“结对编程”。在这种方法中，两名学生合作解决同一个问题。一名学生扮演“驾驶员”的角色，负责编写代码并将设计概念转化为实际的代码实现。与此同时，另一名学生，被称为“观察员”，他审查代码，识别错误，并提供设计、改进建议。在整个会话中，“驾驶员”和“观察员”的角色不断交替。

Pair Programming Lab:

1. The Lab TA will create a pair within the group.
2. Talk to your pair, who is going to be the Driver first.
3. Complete Task 1 with **Pair Programming** (30-45 minutes).
4. Show the TA the results of your works in Task 1.
5. **Switch roles:** the Observer becomes the Driver and vice versa.
6. Complete Task 2 with **Pair Programming** (30-45 minutes).
7. Show the TA the results of your works in Task 2.
8. TA gives grade.

Pair Programming Lab 12 Task 1 – Recursive List Check

- Write a Java function that takes a **List** and a number **n**, and uses **recursion** to output **true** if and only if the List contains the number **n at least 2 times**.
- **Don't** use any loops.
- You may want to consider the helper function trick.
- Test your function intensively with **6 test cases**.
- If you have solved it, try to generalize the problem with "**at least k times**".

Pair Programming Lab 12 Task 2 – Recursive String Check

- Switch role Driver $\leftrightarrow$ Observer.
- Write a Java function that takes two **Strings**: *source* and *target*, and uses **recursion** to output **true** if and only if the string *source* contains a substring *target* **at least 2 times, without overlapping**.
- **Don't** use any loops.
- You may want to consider the helper function trick.
- Test your function intensively with **6 test cases**.
- If you have solved it, try to generalize the problem with "**at least k times**", and also consider the case of counting and allowing *target* to **overlap** in *source*.

This is the end of CPT111-2324 Pair Programming Lab 12 Task Sheet.